PlaneSplit

Control-Plane and Data-Plane Consistency Verification

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Problem Statement #31

- Modern networks rely on two independent layers: Control Plane (Intent) and Data Plane (Physical state).
- Updates to the network are not instantaneous.
- Divergence occurs when the intended routing rules (RIB) do not match the physical hardware rules (FIB).
- This divergence causes packet loss, routing loops, and security blackholes.
- Our Goal: Detect this divergence actively without relying solely on static configuration diffs.

Solution Architecture

- A Pure Software Model built in Python (FastAPI).
- Strict separation of RIB (Control Plane) and FIB (Data Plane) at the routing level.
- Longest Prefix Match (LPM) algorithms resolve next-hops exactly like physical hardware.
- Instead of diffing configs, we inject active simulated probes to trace the actual path.
- WebSocket streaming pushes real-time telemetry to the visualization layer.

Active Boundary Probing

- We simulate packets sent to the network boundaries (e.g., first and last IP of a subnet).
- Control Plane Probe traces the path using the intended RIB rules.
- Data Plane Probe traces the path using the actual FIB hardware rules.
- If paths diverge, the Verification Engine triggers a realtime Alert.
- Handles edge cases like subnet corruption (/25 vs /24) which static diffs might miss.

3D 'Packet Rider' Visualization

- Built with React, Three.js, and React Three Fiber.
- Provides a 'AAA Cyberpunk' aesthetic for maximum presentation impact.
- Green Hologram tracks represent Control Plane intent.
- Red/Blue solid tracks represent the physical Data Plane reality.
- Discrete packet particle explosions provide immediate feedback on success vs. dropped packets.
- Hardware faults trigger dynamic glowing alerts on the 3D Server Racks.

Live Demo Scenarios

- 1. Normal Convergence: Network seamlessly shifts traffic to the AWS ALB.
- 2. Injected Delay: Simulated 2-second propagation delay. Yellow pulsing rack, temporary divergence, auto-recovers.
- 3. Injected Drop: Packet physically drops off the track. Red flashing rack, permanent divergence alert.
- 4. Corrupted Mask: Shows /25 vs /24 misconfiguration caught by boundary probing.

Thank You

Ready for Live Demonstration